

Quiz Website

5-Person Team Plan — 3-Week Sprint

This document contains the full project analysis, team role assignments, week-by-week task breakdown, key deadlines, and feature scope for the CS108 Stanford Quiz Website project. The project is implemented in **Java** using vanilla Servlets and JSPs with a MySQL database backend.

Course	CS108, Stanford University — Object-Oriented Programming (Java)
Project	Quiz Website — take, create, and share quizzes
Team size	5 persons
Timeline	3 weeks (21 days)
Tech stack	Java Servlets, JSPs, MySQL, optional jQuery (client-side)
Grade target	A — requires all required features + extensions

Grading expectations (5-person team)

Required features only → C grade | Required + recommended
→ B grade | Required + recommended + extensions → A grade.
No upper limit — one previous team earned 150% credit.

Team Roles

M1

Member 1

Backend Lead

Quiz engine · Scoring logic · DB schema · Sessions & threading

M2

Member 2

Auth & Users

Login / Register · Password hashing (SHA-256 + salt) · Friends system · Internal mail messages

M3

Member 3

Frontend & UI

JSPs & CSS · Home page · Quiz pages · Semi-professional design

M4

Member 4

Achievements & Admin

6 achievement types · Admin panel · Site statistics · Announcements

M5

Member 5

Extensions & QA

Practice mode · Tags & categories · Improved error handling · Integration & QA

3-Week Sprint Timeline

Week 1 — Foundation

Days 1–7 · Design meeting with teaching staff by Day 4

Who	Task	Details
M1	DB schema design	Users, Quizzes, Questions, Answers, Scores, History, Friends, Achievements tables in MySQL. Finalise and commit by Day 2 so all team members can build against it.
M1	Question type class hierarchy	Abstract Question class with getCorrectAnswers() and grade(String) methods. Subclasses: TextQuestion, FillBlankQuestion, MultipleChoiceQuestion, PictureQuestion. Design for easy extension.
M2	Login & Registration servlets	SHA-256 + salt password hashing (reuse HW4 cracker logic). Store hashed password; compare hash on login.
M2	Session management	HttpSession-based user state; login redirect; logout invalidation.
M3	Global layout & CSS	Shared header/nav JSP fragment, semi-professional stylesheet, typography and colour system.
M3	Login & registration JSPs	Wire up to M2 servlets; form validation messages.
M4	Achievement data model	Achievement enum/class, UserAchievement join table, AchievementService with award-checking hooks.
M4	Admin user table & servlet	isAdmin flag, basic admin landing page, promote-to-admin action.
M5	Category/Tag model	Tag table, QuizTag join table, predefined category list.
M5	web.xml & project scaffold	All servlet mappings, JDBC config, shared DBUtil class for all members.

Week 2 — Core Features

Days 8–14 · Demo meeting with teaching staff by Day 11

Who	Task	Details
M1	Quiz creation servlet	Multi-step form: quiz metadata → add questions of any type → save to DB.
M1	Quiz-taking engine	One-page & multi-page modes, question randomisation option, timer (start time stored in session), answer submission.

Who	Task	Details
M1	Scoring & history	Calculate score + elapsed time, persist to History table, rank by correct answers then by time taken.
M2	Friends system	Send friend request (generates mail message), accept/decline, remove friend, user search by username.
M2	Internal mail servlet	Friend Request, Challenge, Note message types; inbox page with accept/decline actions.
M2	User profile page	Public page with add-friend button, quiz history, earned achievements.
M3	Home page JSP	Announcements, popular/recent quizzes, recent activity, friend feed, achievement icons, messages indicator.
M3	Quiz creation JSP	Dynamic form with JavaScript for adding question blocks; radio buttons for MC; image URL field for picture type.
M3	Quiz summary & results pages	Top scorers table (ranked), user history, comparison with friends, top performers last 24h section.
M4	Achievement triggers	Hook into quiz-taken, quiz-created, top-score events; award logic with deduplication (once earned, never removed).
M4	Admin panel	Announcements CRUD, remove users/quizzes, clear quiz history, promote to admin, site stats (user count, quizzes taken).
M5	Immediate correction mode	Per-question feedback on multi-page quizzes; works with picture-response for flashcard style.
M5	Tag search & category browsing	Search quizzes by tag, filter by category, category listing page.
M5	Integration testing	End-to-end smoke tests across M1–M3 flows; prepare running demo showing contributions from at least 2 members.

Week 3 — Polish, Extensions & Submission

Days 15–21 · Submit zip + README by 9pm Day 21

Who	Task	Details
M1	Multi-answer question type (extension)	Ordered & unordered variants, partial scoring (each blank counts as a separate answer).
M1	Rating system (extension)	Star rating on quiz summary page; re-rank popular quizzes using rating data.
M1	Bug fixing	Scoring edge cases, multi-answer grading, history display.

Who	Task	Details
M2	Cookie-based auto-login (extension)	Remember-me cookie with hashed token; auto-authenticate returning users.
M2	Complete challenge message flow	Sends challenger's best score + quiz link to recipient.
M2	Non-registered access (extension)	Allow guests to view announcements, quiz list, and quiz summary pages before logging in.
M3	UI polish pass	Achievement icons/tooltips (mouse-over description), responsive layouts, consistent typography, nav polish.
M3	Quiz editing page (extension)	Allow quiz owner to modify questions after creation.
M3	README.html	Document all features, team members, completed extensions, and anything skipped.
M4	Reporting system (extension)	Flag quiz as inappropriate; admin review queue with approve/remove actions.
M4	Additional achievements	Practice Makes Perfect, I am the Greatest (with deduplication); custom colourful icons for all 6+ achievements.
M4	Admin stats page	User count, quizzes taken, top quizzes, recent signups chart.
M5	Practice mode (required extension)	Repeat loop, track per-question correct count, remove question after 3 correct answers, terminate when all cleared.
M5	Improved error handling (extension)	Friendly error JSPs for bad inputs, no raw 404s or server errors.
M5	Final QA & submission	Full click-through of every page; verify no raw server errors; zip code; submit; prepare for grading meeting on Day 22.

Key Deadlines

When	Deadline	Notes
Week 1 · Day 4	Design meeting with teaching staff	Meet Dr. Young or a TA to discuss design & task assignments before this date.
Week 2 · Day 11	Demo meeting — running code from 2+ members	Show integrated code with contributions from at least two team members. This is the early-integration checkpoint.
Week 3 · Day 21	Code submission by 9pm	Zip all code + README.html (or README.doc). Only one team member submits the zip; others submit a text file naming the submitter.
Day 22 (Saturday)	Grading meeting with TA	Afternoon grading session — scheduling details provided closer to the date.
Day 23 (Monday)	Team evaluation email by midnight	Email to patrick.young@stanford.edu listing each member, what they did, and estimated percentage contribution.
CRITICAL	Late penalty — draconian	Projects are penalised per minute late. A project submitted just 3 hours late receives a ZERO. Submit early.

Feature Scope (5-Person Team)

REQUIRED (needed for any passing grade)

- 4 question types: Question-Response, Fill-in-the-Blank, Multiple Choice (radio buttons), Picture-Response
- Quiz options: randomise questions, one-page vs multi-page, immediate correction mode
- Practice mode toggle on quiz (required for 5-person teams)
- Scoring: correct-answer count + elapsed time; rank by score then time
- User accounts with SHA-256 + salt hashed passwords
- Friends system: two-step confirmation (request + accept), remove friend, user search
- Internal mail: Friend Request, Challenge (with best score + link), Note
- Quiz history for every user (on profile page and abbreviated on home page)
- 6+ achievements: Amateur/Prolific/Prodigious Author, Quiz Machine, I am the Greatest, Practice Makes Perfect
- Admin panel: announcements, remove users/quizzes, clear history, promote admins, site stats
- Semi-professional page design (required for 5-person teams)
- Multiple simultaneous users via HttpSession (no thread locking required)

- Error checking: no raw 404s or server error pages from normal navigation

EXTENSIONS (needed for A grade — 5-person teams must do these)

- Practice mode algorithm: repeat loop, remove question after 3 correct, terminate when all cleared
- Tags & categories: tag search, category browse, predefined category list
- Rating & review system: star ratings affect popular-quizzes ranking
- Cookie-based auto-login (remember-me token)
- Non-registered access: guests can view announcements, quiz list, summary pages
- Quiz editing after creation (owner only)
- Reporting / inappropriate-flag system with admin review queue
- Multi-answer question type: ordered & unordered variants, partial scoring
- Improved form error handling: friendly messages for bad inputs
- Custom achievement icons with mouse-over tooltip descriptions
- Multiple-choice with multiple answers question type
- XML quiz loading format

OOP Architecture & Implementation Notes

1 Question class hierarchy is the core OOP design challenge

Create an abstract Question class with getCorrectAnswers() and grade(String input) methods. Subclass it for TextQuestion, FillBlankQuestion, MultipleChoiceQuestion, and PictureQuestion. This design means adding new question types in extensions (multi-answer, matching) requires zero changes to the quiz engine — only a new subclass.

2 DB schema is the shared critical dependency

Member 1 must finalise and commit the MySQL schema by Day 2 of Week 1 so all other members can build against it. A late schema change in Week 2 will break multiple people's work simultaneously. Agree on column names and types early.

3 Integration risk is the #1 failure mode

The handout explicitly warns that failing teams did not integrate early. By Day 11 (the demo meeting), all five work streams must be talking to the same database. Do not allow members to work in isolated silos until Week 3.

4 Technology restrictions — read carefully

Vanilla Servlets and JSPs only — no Spring, no Hibernate, no other server-side frameworks. All data in MySQL — no text files except web.xml and those explicitly described. jQuery is permitted for client-side work. Absolute image URLs for picture-response questions (no file uploads needed).

5 Password security (reuse HW4 cracker logic)

On registration: generate a random salt, hash (salt + password) with SHA-256, store both hash and salt. On login: retrieve salt, hash (salt + entered password), compare with stored hash. Never store the plaintext password.

6 Scoring rank order

Rank top scorers first by number of correct answers (descending), then by time taken (ascending). A user with 10/10 in 10 minutes ranks above a user with 9/10 in 1 minute. Multi-answer questions: each blank = 1 point.

7 Session handling

Use HttpSession for all user-specific state (logged-in user, quiz in progress, timer start). Multiple users can visit simultaneously — the spec says you do NOT need thread-level locking since users never access the same data at the same time.

8 Immediate correction + practice mode interaction

Immediate correction applies to multi-page quizzes and gives per-question feedback. Practice mode is separate: scores are not recorded, questions repeat, and a question is removed from rotation after 3 correct answers.